Chapter 21 · Semester 3

SOUND WAVES

Notes by Cikgu Suhaili

21 worked examples 4 simulations Exercises with answers

21.1 Propagation of sound waves

- A sound wave is a **longitudinal wave** that needs a medium (solid, liquid or gas) to travel.
- The source of a sound wave is a vibrating object.
- The sound wave travels as a series of **compressions and rarefactions** of the medium.
- Sound travels slowest in gases, faster in liquids and fastest in solids. In a gas the speed depends on temperature: in air, $v \approx (331 + 0.606\theta)$ m s^{-1}, θ in °C.

Material (20 °C)	Speed of sound / m s^{-1}
Air	343
Air (0 °C)	331
Helium	1005
Water	1440
Sea water	1560
Glass	≈ 4500
Iron and steel	≈ 5000
Aluminium	≈ 5100

How a tuning fork produces sound

- When the prongs move outwards, they push the air in front of them together. The pressure rises slightly: a **compression** (condensation).
- When the prongs move inwards, the air spreads out. The pressure falls slightly below normal: a **rarefaction**.
- As the fork keeps vibrating, compressions and rarefactions are produced one after another and spread away from it.

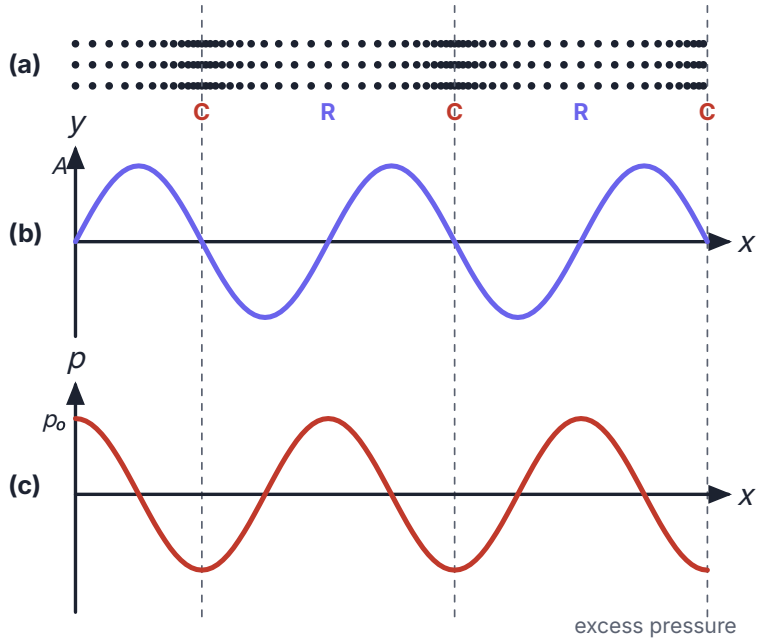


Figure 21.1: (a) the air layers at one instant (C = compression, R = rarefaction); (b) displacement y of the air particles against distance x; (c) excess pressure p against distance x.

Displacement and pressure

Displacement of the air particles

$$y = y_0 \sin(\omega t - kx)$$

Pressure variation (excess pressure)

$$p = p_0 \sin\left(\omega t - kx + \frac{\pi}{2}\right)$$

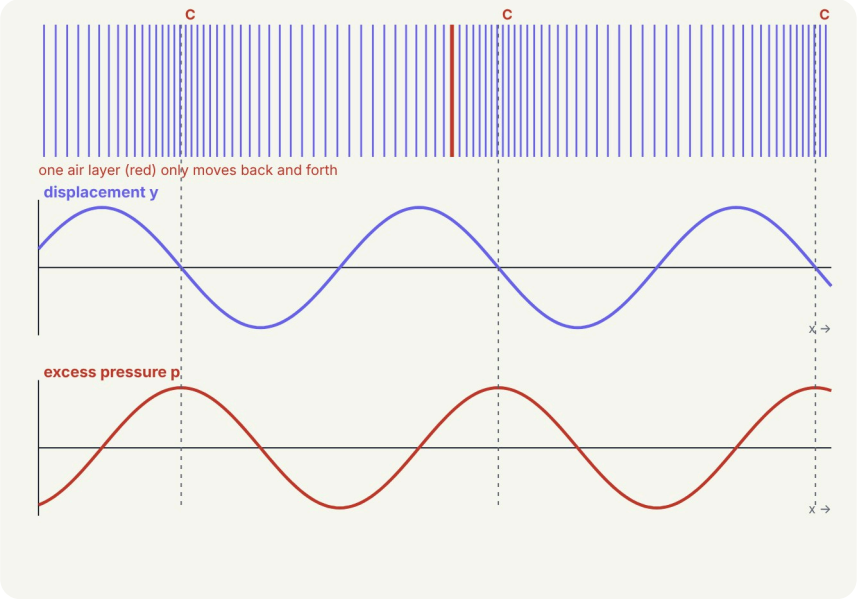
- At a compression or a rarefaction, the particles have **zero displacement** (they pass through their equilibrium position at maximum speed). Here the pressure is **maximum** (compression, above p_{atm}) or **minimum** (rarefaction, below p_{atm}).
- Where the displacement is maximum ($y = \pm y_0$), the particles are momentarily at rest, their spacing is normal, and the excess pressure is zero.
- So the pressure wave and the displacement wave are **π/2 rad (a quarter cycle) out of phase**: the pressure wave leads the displacement wave by π/2.

phase difference between p and y: $\Delta\varphi = \frac{\pi}{2}$ rad

p_0 = maximum excess pressure = $p_{max} - p_{atm}$

Simulation — Sound wave: displacement and pressure

Watch the air layers. Each layer only vibrates back and forth about its own position. The compressions (dark) and rarefactions (light) move along. Compare the displacement graph with the pressure graph: they are a quarter cycle apart.



This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

Example 21.1.1

A sound wave in air is represented by $y = 4.0 \times 10^{-8} \sin(1200\pi t - 3.5\pi x)$ m. Find (a) the amplitude, (b) the wavelength, (c) the frequency, (d) the speed of sound in air.

Solution

Compare with $y = y_0 \sin(\omega t - kx)$.

(a) $y_0 = \mathbf{4.0 \times 10^{-8}\ m}$

(b) $k = 3.5\pi \Rightarrow \lambda = 2\pi/k = \mathbf{0.571\ m}$

(c) $\omega = 1200\pi \Rightarrow f = \omega/2\pi = \mathbf{600\ Hz}$

(d) $v = f\lambda = 600 \times 0.571 = \mathbf{343\ m\,s^{-1}}$

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Describe the propagation of sound waves in air. [2]

Solution

- Sound is a series of compressions and rarefactions (pressure variations) that travel through the air.
- Sound waves travel in air as longitudinal waves: the air particles vibrate parallel to the direction of propagation.

21.2 Sources of sound

The source of any sound is a vibrating object, e.g. musical instruments (guitar, violin, flute, trumpet). The vibrations are standing waves on strings or in air columns.

Stationary waves on a stretched string

- When a stretched string is plucked, a transverse progressive wave travels to the fixed ends and is reflected.
- The incident and reflected waves (same frequency and amplitude) superpose to form a **stationary transverse wave**.
- Both fixed ends are **nodes**. The vibrating string makes the air vibrate at the same frequency, producing sound.

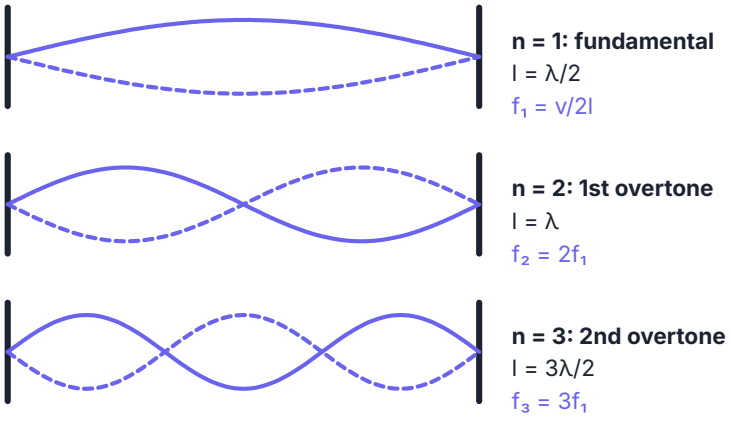


Figure 21.2: the first three modes of vibration of a string fixed at both ends.

$$f_n = \frac{nv}{2l} = nf_1, \quad n = 1, 2, 3, \ldots \qquad v = \sqrt{\frac{T}{\mu}}$$

n = harmonic number (number of loops); l = length of string; T = tension; μ = mass per unit length (kg m^{-1}); all harmonics are present

Term	Meaning
Fundamental frequency	the lowest natural frequency of a vibrating system
Harmonics	frequencies that are whole-number multiples of the fundamental frequency
Overtones	the frequencies above the fundamental that the system can produce: the first overtone is the next one after the fundamental

Example 21.2.1

A string vibrates in five segments at a frequency of 460 Hz. (a) What is its fundamental frequency? (b) What frequency makes it vibrate in three segments?

Solution

(a) five segments: $n = 5$; $f_5 = 5f_1 \Rightarrow f_1 = 460/5 =$ **92 Hz**

(b) $f_3 = 3(92) =$ **276 Hz**

Example 21.2.2

A steel piano wire 0.400 m long has a mass of 3.00 g and is under a tension of 800 N. Find (a) the frequency of its fundamental mode, (b) the highest harmonic that could be heard by a person who can hear frequencies up to 10 000 Hz.

Solution

$\mu = m/l = 3.00 \times 10^{-3}/0.400 = 7.5 \times 10^{-3}$ kg m^{-1}

(a) $f_1 = \frac{1}{2l}\sqrt{\frac{T}{\mu}} = \frac{1}{2(0.400)}\sqrt{\frac{800}{7.5 \times 10^{-3}}} =$ **408 Hz**

(b) $n = 10\,000/408 = 24.5$, so the highest harmonic heard is the **24th**.

STPM 2021 · Section A

A string is fixed at both ends and plucked in the middle. The length, mass and tension of the string are 2.0 m, 8.0 g and 1.2 kN respectively. What is the fundamental frequency of the string?

A. 4.3 Hz

B. 68 Hz

C. 140 Hz

D. 270 Hz

Answer

> **Answer: C**
>
> $\mu = 0.0080/2.0 = 4.0 \times 10^{-3}$ kg m^{-1}; $v = \sqrt{(1200/4.0 \times 10^{-3})} = 548$ m s^{-1}; $f_1 = v/2L = 548/4.0 = 137$ Hz ≈ 140 Hz.

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A string of length L is stretched between two fixed points with tension T. Sketch the shape of the string for the second overtone, and find the distance between two consecutive nodes in terms of the fundamental wavelength. [4]

Solution

> Second overtone = third harmonic ($n = 3$): the string forms **3 loops**.
>
> $L = 3(\lambda_3/2) \Rightarrow$ distance between nodes $= \lambda_3/2 = L/3$
>
> fundamental: $\lambda_1 = 2L \Rightarrow L = \lambda_1/2$, so the distance = **$\lambda_1/6$**

Stationary waves in air columns

- A sound wave sent into a pipe is reflected at the end. The incident and reflected waves superpose to form a **stationary longitudinal wave**.
- **Resonance** occurs, and a loud sound is heard, when the frequency of the source equals a natural frequency of the air column.
- A **closed end** is always a **displacement node** (the air cannot move). An **open end** is a **displacement antinode**.

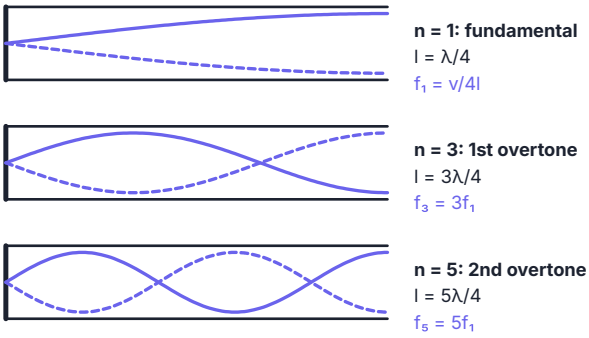


Figure 21.3: closed pipe (one end closed). Only odd harmonics.

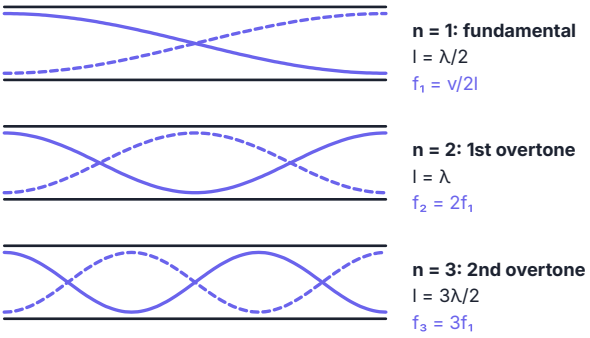


Figure 21.4: open pipe (both ends open). All harmonics.

	Closed pipe	Open pipe and string
Fundamental	$f_1 = v/4l$	$f_1 = v/2l$
General	$f_n = nv/4l$, n = 1, 3, 5, ... (odd harmonics only)	$f_n = nv/2l$, n = 1, 2, 3, ...
1st overtone	$3f_1$	$2f_1$
2nd overtone	$5f_1$	$3f_1$

End correction

End correction, c

is **the small distance between the open end of a pipe and the antinode just outside it**. The air at the open end is free to move, so its vibration extends slightly beyond the end of the pipe.



Figure 21.5: the antinode forms a distance c beyond each open end.

Closed pipe	Open pipe
$l + c = \lambda/4 \Rightarrow f_1 = \frac{v}{4(l + c)}$	$l + 2c = \lambda/2 \Rightarrow f_1 = \frac{v}{2(l + 2c)}$
$f_n = \frac{nv}{4(l + c)}$, n = 1, 3, 5, ...	$f_n = \frac{nv}{2(l + 2c)}$, n = 1, 2, 3, ...

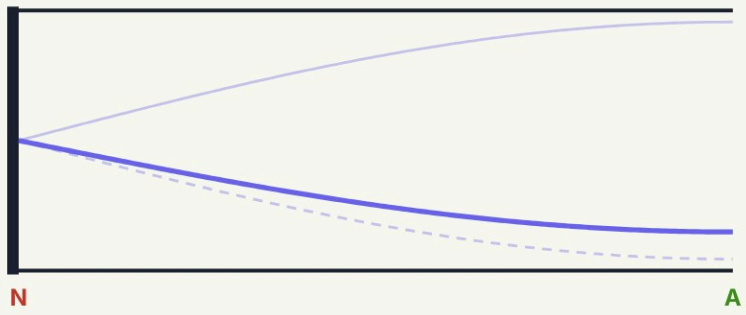
Teacher's note

To find the end correction experimentally, use a resonance tube: find the first two resonance lengths l_1 and l_2 for the same tuning fork. Then $l_2 - l_1 = \lambda/2$ (the end correction cancels), and $c = \lambda/4 - l_1$.

Simulation

Harmonics in strings and pipes

Choose a string, an open pipe or a closed pipe, then step through the harmonics. Change the length and switch on the end correction. A closed pipe only has odd harmonics.



displacement of the air (drawn sideways)

harmonic	overtone	λ	frequency (v = 340 m/s)
n = 1 (fundamental)	**—**	**2.000 m**	**170 Hz**

formula

f = nv/4(l + c)

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

Example 21.2.3

A pipe is 45.0 cm long. Find the fundamental frequency and the first three overtones (a) if the pipe is open at both ends, (b) if it is closed at one end. (c) For each case, find the highest harmonic that can be heard by a person who hears 20 Hz to 20 000 Hz. (v = 344 m s^{-1})

Solution

(a) open: $f_1 = v/2l$ = 344/0.90 = **382 Hz**; overtones f_2, f_3, f_4 = **764, 1147, 1529 Hz**

(b) closed: $f_1 = v/4l$ = 344/1.80 = **191 Hz**; overtones f_3, f_5, f_7 = **573, 956, 1338 Hz**

(c) open: n = 20 000(0.90)/344 = 52.3 ⇒ **52nd harmonic**. Closed: n = 20 000(1.80)/344 = 104.7, but only odd harmonics exist ⇒ **103rd harmonic**.

Example 21.2.4

A pipe 90 cm long is open at both ends. How long must a pipe closed at one end be to have the same fundamental frequency?

Solution

$$\frac{v}{4l_c} = \frac{v}{2l_o} \Rightarrow l_c = \frac{l_o}{2} = \textbf{45 cm}$$

Example 21.2.5

A tube of air 1.5 m long is closed at one end. It sustains a standing wave at its third harmonic. What is the distance between a node and the adjacent antinode? (v = 331 m s^{-1})

Solution

Third harmonic: $l = 3\lambda/4 \Rightarrow \lambda = 4(1.5)/3 = 2.0$ m

node to adjacent antinode = $\lambda/4$ = **0.5 m**

Example 21.2.6

A pipe closed at one end produces a fundamental note of 300 Hz. The end correction is 8.00 mm and the speed of sound is 330 m s^{-1}. Calculate the length of the pipe.

Solution

$f_1 = \dfrac{v}{4(l + c)} \Rightarrow l + c = \dfrac{330}{4(300)} = 0.275$ m

$l = 0.275 - 0.008 =$ **0.267 m**

Exercises: strings and pipes

1. The distance between the first and the fourth antinodes of a stationary wave on a string is 15 cm. What is the speed of the wave if it vibrates at 50 Hz?

   Answer

   $3 \times \lambda/2 = 15$ cm $\Rightarrow \lambda = 0.10$ m; $v = 5$ m s^{-1}

2. A string 2.0 m long is fixed at both ends. At 85.0 Hz a standing wave with five loops forms. Find (a) the wavelength, (b) the wave speed, (c) the fundamental frequency.

   Answer

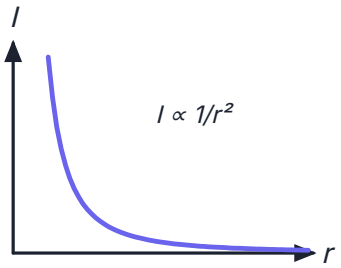
   (a) 0.8 m (b) 68 m s^{-1} (c) 17 Hz

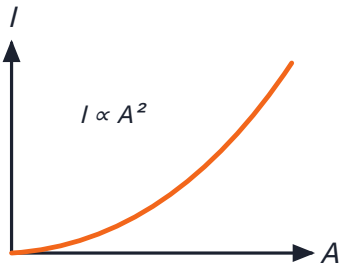
21.3 Intensity level of sound

Intensity of sound

is **the rate of energy transported per unit area perpendicular to the direction of propagation**: $I = \dfrac{E}{tA} = \dfrac{P}{A}$, unit W m^{-2}.

For a point source radiating uniformly, $A = 4\pi r^2$, so $\dfrac{I_2}{I_1} = \dfrac{r_1^2}{r_2^2}$. The intensity (and so the loudness) depends on the amplitude, $I \propto A^2$, and on the distance, $I \propto 1/r^2$.





Example 21.3.1

A loudspeaker radiates sound uniformly in all directions. At 3 m the intensity is 0.4 W m^{-2}. Find the power of the loudspeaker and the intensity at 5 m.

Solution

$P = I(4\pi r^2) = 0.4 \times 4\pi(3)^2 =$ **45.2 W**

$I_2 = I_1(r_1/r_2)^2 = 0.4(3/5)^2 =$ **0.144 W m^{-2}**

Intensity level

- **Threshold of hearing**: the lowest intensity a normal ear can detect, $I_0 = 1.0 \times 10^{-12}$ W m^{-2}.
- **Threshold of pain**: about 1 W m^{-2}.
- Our sense of loudness follows the logarithm of the intensity, so a logarithmic scale is used.

Intensity level of sound, β

is **a logarithmic measure of the intensity of a sound compared with the threshold of hearing**.

$$\beta = 10 \log_{10} \frac{I}{I_0}$$

unit: decibel (dB). $I_0 = 1.0 \times 10^{-12}$ W m^{-2}. An increase of 10 dB means the intensity is 10 times greater; +3 dB means the intensity doubles.

Sound	Intensity level / dB
threshold of hearing	0
whisper	20
conversation, classroom	60
siren, rock concert	100
jet aeroplane, threshold of pain	120

Example 21.3.2

The intensity of a sound wave is 1 W m^{-2}. Calculate its intensity level.

Solution

$\beta = 10 \log_{10} (1/10^{-12}) = 10 \times 12 =$ **120 dB**

Example 21.3.3

The intensity level of one sound is 80 dB and of another is 20 dB. Find the ratio of their intensities.

Solution

$I_1/I_0 = 10^8$, $I_2/I_0 = 10^2 \Rightarrow I_1/I_2 =$ **10^6**

Shortcut: $\beta_1 - \beta_2 = 10 \log (I_1/I_2) \Rightarrow 60 = 10 \log (I_1/I_2)$

STPM 2022 · Section A

The sound intensity level of 88.0 dB is detected 30.0 cm from a point source. What is the energy received at that point in one second? [$I_0 = 1.00 \times 10^{-12}$ W m^{-2}]

A. 1.19×10^{-3} J

B. 7.14×10^{-4} J

C. 5.58×10^{-3} J

D. 1.78×10^{-3} J

Answer

Answer: B

$I = 10^{8.8} \times 10^{-12} = 6.31 \times 10^{-4}$ W m^{-2}; $P = I \times 4\pi r^2 = 6.31 \times 10^{-4} \times 4\pi(0.300)^2 = 7.14 \times 10^{-4}$ W, i.e. 7.14×10^{-4} J each second.

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A bomb blasts 12 m from a soldier. The area of his eardrum is 25 mm^2 and the sound energy it receives every second is 1.0×10^{-6} W. [$I_0 = 10^{-12}$ W m^{-2}] (a) Find the sound intensity level at the eardrum. [3] (b) Find the change in level if the soldier is 17 m from the blast. [3] (c) If the level increases by 3 dB, what happens to the intensity? [1]

Solution

(a) $I = P/A = 1.0 \times 10^{-6}/25 \times 10^{-6} = 0.040$ W m^{-2}; $\beta = 10 \log (0.040/10^{-12})$ = **106 dB**

(b) $I_2 = 0.040(12/17)^2 = 0.0199$ W m^{-2}; $\beta_2 = 103.0$ dB ⇒ a **decrease of 3.0 dB**

(c) $3 = 10 \log (I_2/I_1) \Rightarrow I_2/I_1 = 10^{0.3} \approx 2$: the intensity **doubles**.

21.4 Beats

Beats

are **the periodic variation in amplitude (loudness) at a point, caused by the superposition of two sound waves of slightly different frequencies**.

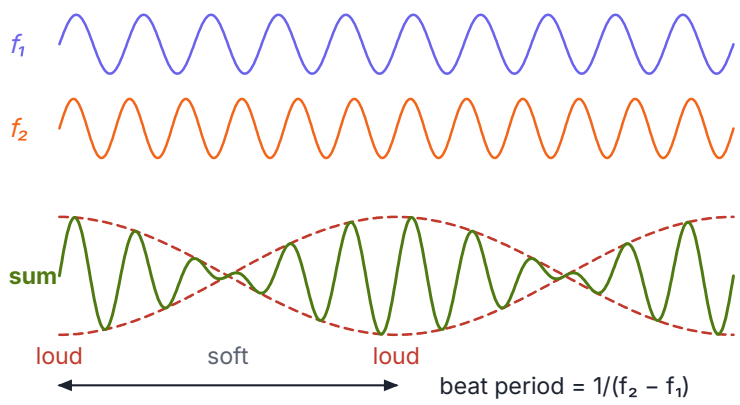


Figure 21.6: two waves of 50 Hz and 60 Hz and their sum. The loudness rises and falls 10 times per second.

- Where the two waves are in phase, they interfere constructively: the amplitude is maximum and a loud sound is heard.
- Where they are in antiphase, they interfere destructively: the amplitude is minimum and little or no sound is heard.
- The number of times per second the loudness rises and falls is the **beat frequency**.

Derivation of the beat frequency

$y_1 = A \sin 2\pi f_1 t, \quad y_2 = A \sin 2\pi f_2 t \quad (f_1 > f_2)$

$y = y_1 + y_2$, with $\sin P + \sin Q = 2 \cos \frac{P - Q}{2} \sin \frac{P + Q}{2}$

$y = [2A \cos 2\pi(\frac{f_1 - f_2}{2})t] \sin 2\pi(\frac{f_1 + f_2}{2})t$

The sound heard has frequency $(f_1 + f_2)/2$. Its amplitude $2A \cos 2\pi(\frac{f_1 - f_2}{2})t$ varies slowly.

The loudness is maximum whenever the cosine is +1 **or** −1, i.e. twice in each cycle of the cosine. So the beat frequency is twice $(f_1 - f_2)/2$.

$$f_{\text{beat}} = f_1 - f_2 = \frac{v}{\lambda_1} - \frac{v}{\lambda_2}$$

beat period $T_{\text{beat}} = 1/f_{\text{beat}}$. For 50 Hz and 60 Hz: 10 beats per second, $T_{\text{beat}} = 0.10$ s.

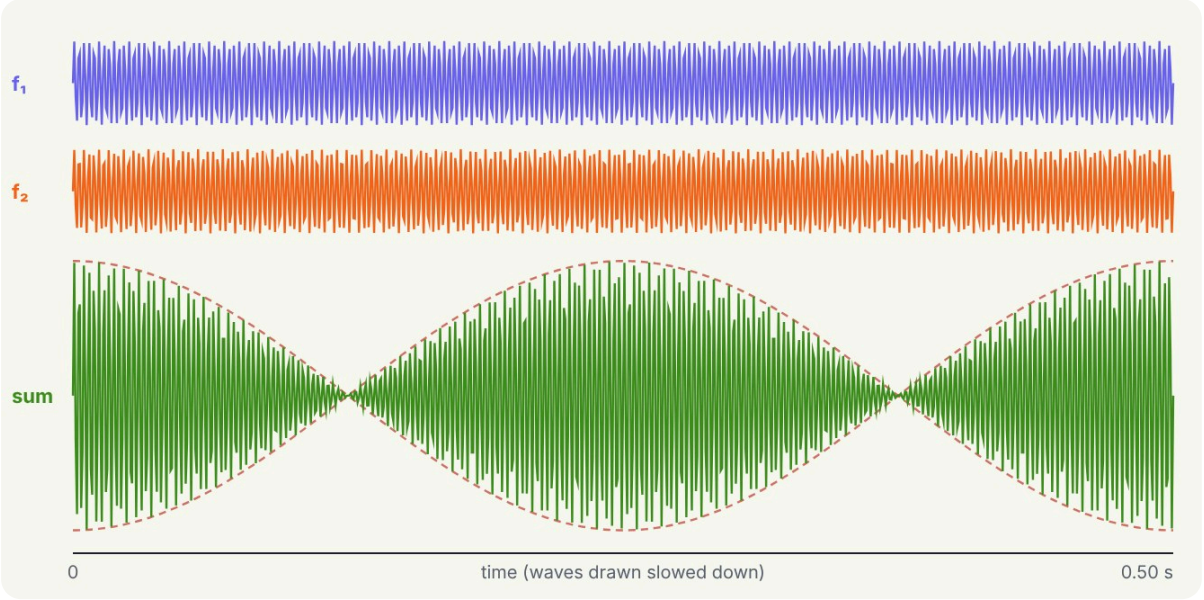
Teacher's note

Beats are used to tune instruments: the string is adjusted until the beats with a reference tuning fork slow down and disappear ($f_{\text{beat}} = 0$).

Simulation

Beats

Set two frequencies that are close together. The green curve is their sum: its amplitude rises and falls at the beat frequency. Press Listen to hear the beats (use earphones or a speaker).



beat frequency $\lvert f_1 - f_2 \rvert$	beat period	frequency heard $(f_1 + f_2)/2$
4 Hz	**0.25 s**	**442.0 Hz**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

Example 21.4.1

Two sound waves of wavelengths 3.00 m and 2.50 m superpose. Both travel at 330 m s^{-1}. How many beats are produced each second?

Solution

$$f_{\text{beat}} = \frac{330}{2.50} - \frac{330}{3.00} = 132 - 110 = \textbf{22 Hz}, \text{ i.e. 22 beats per second}$$

STPM 2022 · Section A

A musician plucks the first two strings of a ukulele together and hears three beats per second. The first string is tuned to 350 Hz. The musician tightens the screw to stretch the second string, plucks the two strings again and hears two beats per second. What was the frequency of the second string before it was stretched?

A. 345 Hz

B. 347 Hz

C. 351 Hz

D. 353 Hz

Answer

Answer: B

At first $|350 - f_2| = 3$, so f_2 = 347 Hz or 353 Hz. Tightening increases the tension, so f_2 increases. The beat frequency falls from 3 to 2, so f_2 moved closer to 350 Hz: it was **347 Hz** (now 348 Hz).

21.5 Doppler effect

Doppler effect

is **the change in the observed frequency of a wave because of the relative motion between the source and the observer**.

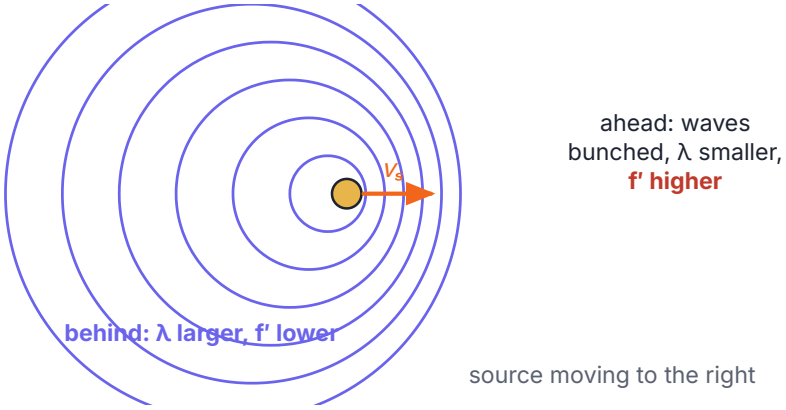


Figure 21.7: a moving source. The wavefronts are closer together in front of the source and further apart behind it.

- Source at rest: every observer hears the source frequency f_s.
- Source moving **towards** the observer: the wavelength reaching the observer is shorter, so the observed frequency is **higher** ($f' > f_s$).
- Source moving **away**: the wavelength is longer, so the observed frequency is **lower** ($f' < f_s$).

Derivation

Moving source (speed u_s, towards the observer): in one period $T = 1/f$, the source moves $u_s T$ while the wave moves vT. The wavelength ahead is $\lambda' = (v - u_s)/f$.

$$f' = v/\lambda' = \frac{v}{v - u_s} f \quad \text{(away: } v + u_s\text{)}$$

Moving observer (speed u_o, towards the source): the waves pass the observer at relative speed $v + u_o$; the wavelength is unchanged, $\lambda = v/f$.

$$f' = \frac{v + u_o}{\lambda} = \frac{v + u_o}{v} f \quad \text{(away: } v - u_o\text{)}$$

$$f' = \frac{v \pm u_o}{v \mp u_s} f$$

v = speed of sound; u_o, u_s = speeds of observer and source along the line joining them. Upper signs: moving **towards** each other (f' increases). Lower signs: moving **away** (f' decreases).

Teacher's note

Decide the sign from the physics, not from memory: "towards" must make f' bigger (bigger numerator or smaller denominator); "away" must make f' smaller.

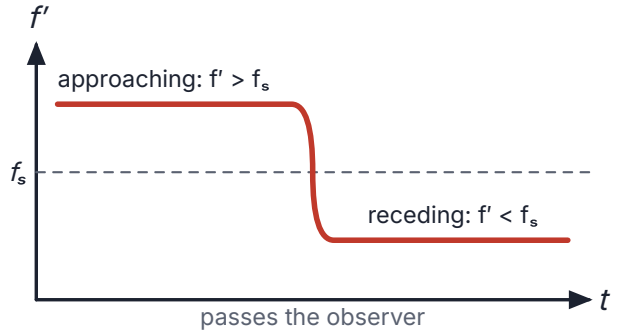
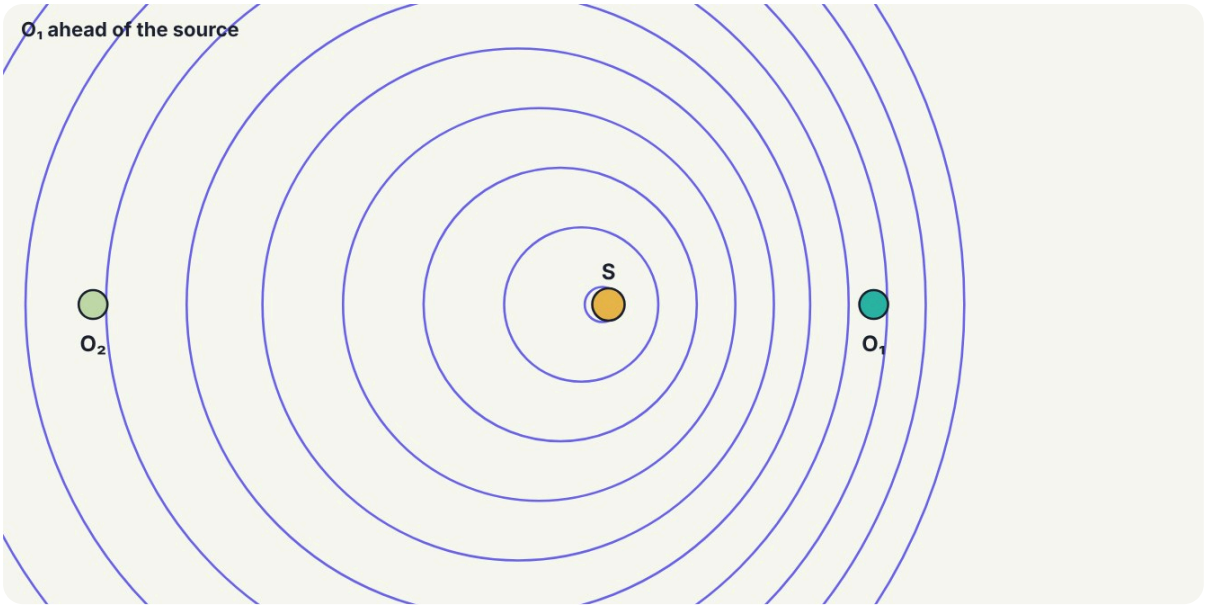


Figure 21.8: the frequency heard by a stationary observer as a source approaches, passes and moves away at constant speed.

- While the source approaches, the observed frequency is constant and higher than f_s; it drops to f_s as the source passes; it is then constant and lower while the source moves away.
- The **loudness** does change: as the source comes closer, r decreases and $I \propto 1/r^2$ increases. Do not confuse loudness (intensity) with pitch (frequency).

Simulation Doppler effect

Move the source, or the observer, and watch the wavefronts. Compare the frequency heard by the observer in front of the source with the one behind it.



source frequency f_s	O_1 hears	O_2 (behind, at rest) hears
400 Hz	**618 Hz**	**296 Hz**

formula

$f' = f(v \pm u_o)/(v \mp u_s)$, $v = 340$ m/s

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

Example 21.5.1

A factory whistle emits sound at 900 Hz. What frequency is heard by a person in a car travelling at 70 km h^{-1} (a) away from the whistle, (b) towards it? (v = 340 m s^{-1})

Solution

u_o = 70 km h^{-1} = 19.4 m s^{-1}

(a) $f' = \dfrac{340 - 19.4}{340}(900) =$ **849 Hz**

(b) $f' = \dfrac{340 + 19.4}{340}(900) =$ **951 Hz**

Example 21.5.2

An observer moves along the line joining two identical sources, each of frequency 500 Hz. The observer hears 4.0 beats per second. Find the speed of the observer. (v = 340 m s^{-1})

Solution

Moving towards one source: $f_1' = \dfrac{v + u_o}{v} f$ (higher). Moving away from the other: $f_2' = \dfrac{v - u_o}{v} f$ (lower).

$$f_{beat} = f_1' - f_2' = \frac{2fu_o}{v} \Rightarrow 4.0 = \frac{2(500)u_o}{340}$$

u_o = **1.36 m s^{-1}**

Example 21.5.3

Two identical sources S_1 and S_2, each of frequency 500 Hz, move towards a stationary observer at 20 m s^{-1} and 23 m s^{-1}. Find the beat frequency heard. (v = 340 m s^{-1})

Solution

$$f_1' = 500 \times \frac{340}{340 - 20} = 531.25 \text{ Hz}; \quad f_2' = 500 \times \frac{340}{340 - 23} = 536.28 \text{ Hz}$$

f_{beat} = 536.28 − 531.25 = **5.03 Hz**

STPM 2022

An ambulance moves at 42 m s^{-1} and sounds its siren at 960 Hz. A bus in front of the ambulance stops at the roadside. [v = 330 m s^{-1}] (i) Find the frequency of the siren heard by the passengers in the bus. [2] (ii) The sound is reflected by the bus. Find the frequency and the wavelength of the reflected sound heard by the ambulance driver. [2] (iii) What change in frequency do the passengers hear after the ambulance overtakes the bus? [1]

Solution

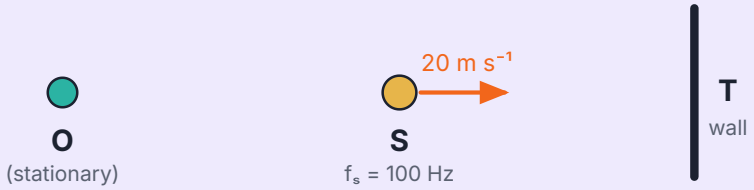
(i) $f' = 960 \times \dfrac{330}{330 - 42} =$ **1100 Hz**

(ii) The bus reflects 1100 Hz as a stationary source; the driver moves towards it: $f' = 1100 \times \dfrac{330 + 42}{330} =$ **1240 Hz**. The reflected wave in the air has $\lambda = v/f' = 330/1100 =$ **0.300 m** (a moving observer changes the frequency heard, not the wavelength in the air).

(iii) The frequency heard **decreases** (the siren now moves away).

Exercise: Doppler effect and beats

1. O is a stationary observer. Source S moves away from O towards a wall T at 20 m s^{-1}; its frequency is 100 Hz. (v = 340 m s^{-1}) Find (a) the wavelength in front of and behind the source (ignore the wall), (b) the frequency O hears directly from S, (c) the frequency O hears after reflection from T, (d) the beat frequency O hears.



Answer

(a) in front: (340 − 20)/100 = 3.2 m; behind: (340 + 20)/100 = 3.6 m (b) 100 × 340/360 = 94 Hz (c) 100 × 340/320 = 106 Hz (d) 106 − 94 = 12 Hz